

Chapter 15

Untangling the IP Web: Ownership, Rights, and Strategies in University–Industry Collaboration



Stephen J. Susalka and Hannah Dvorak Carbone

15.1 Introduction

Intellectual property (IP) is a legal construct designed to protect intangible products of human creativity and ingenuity. IP includes a number of protection mechanisms (Wilson, 2008). The main value of IP is as a source of competitive advantage to its owner; generally speaking, an IP owner has the right to exclude others from making use of the IP in question, blocking a potential direct competitor. Types of IP protection include patents, copyrights, trademarks, and trade secrets. In university-industry (UI) partnerships, patents and copyrights are generally the most relevant forms of IP, as explained below.

Patents, in the United States, come in three kinds: utility, plant, and design. Utility patents are by far the most common kind of patent, and protect inventions, provided that those inventions are useful, novel, and nonobvious with respect to the so-called prior art—that which is already known in the field of the invention at the time of submission of a patent application. Plant patents cover new asexually reproduced varieties of plants. Design patents protect the ornamental appearance of an object. In most cases, patents (and pending patent applications) are published,

The authors' contributions to this chapter are their own views and do not necessarily represent the position of their employers.

S. J. Susalka
AUTM, Washington, DC, USA
e-mail: ssusalka@autm.net

H. D. Carbone (✉)
California Institute of Technology, Pasadena, CA, USA
e-mail: hannah@caltech.edu

© The Author(s), under exclusive license to Springer Nature
Switzerland AG 2025

R. Hall, A. Boccanfuso (eds.), *University-Industry Collaboration*, International
Series in Operations Research & Management Science 369,
https://doi.org/10.1007/978-3-031-94913-5_15

making the invention, plant, or design known to the public. Patents have a defined term—20 years from the earliest nonprovisional priority date, for most utility patents—and after the expiration of the patent, the subject matter of the patent is in the public domain and can be used by anyone without compensation to the original patent owner. In many cases, patentable inventions are a significant outcome of the research performed under UI collaborations, and the protections offered by patent law can be a source of commercial competitive advantage for the industry partner.

Copyright covers tangible expressions of creative work (publications, movies, etc.) and protects against the unauthorized reproduction, display, performance, or modification of that work. In UI collaborations, copyright as a form of IP protection is relevant primarily with respect to software; once software code is written, it is considered a copyrightable creative work. (The functionality of software can also be covered by a utility patent.) While other deliverables under a UI collaboration may also be copyrighted or copyrightable (e.g., technical reports, scientific publications), the protection by copyright of the expression of the information in question is of less commercial value than the information itself.

Trademarks identify the origin of goods and services and protect against others from using confusing or deceptive marks. For example, using a logo very similar to the Starbucks logo to sell one's own brand of coffee could deceive a customer into thinking they were buying authentic Starbucks coffee, so this would constitute trademark infringement. Trademarks rarely play a role in UI partnerships; if anything, most universities have guidelines around the use of their names, logos, mascots, or other trademarks that preclude their use in a way that might signal the university's endorsement of a commercial entity, activity, or product.

Trade secrets may be held by an individual or organization when the details of a particular process, or constituents of a particular product, are kept securely hidden from public access. In principle, a trade secret can be kept indefinitely. However, a trade secret does not protect against independent invention or reverse engineering of the covered matter. Trade secrets are commonly used in industry, but are rarely held by universities, whose mission to advance scientific knowledge mandates the publication of discoveries and inventions.

Intellectual property may be one of the most valuable outcomes of UI collaboration—and may often be the most complex and highly negotiated aspect.

This is especially true in research-focused collaborations where new IP is the expected result of the collaboration (if not the primary endpoint). While the creation and protection of IP is of interest to both the university and industry, there are often differing expectations and misalignments in those collaborations. This is generally caused by the fundamental differences between the two types of organizations: universities are generally nonprofit institutions focused on education and the advancement of scientific and other knowledge, whereas companies are generally for-profit entities focused on creating shareholder value through the production and sale of goods and services into the marketplace.

IP terms, therefore, are a common obstacle in the negotiation of agreements governing UI collaborations. The existence of protectable IP may be critical to translating the research results of UI collaboration to a product or service that makes it to

the marketplace. However, the university partner and the industry partner may have conflicting interests regarding the management of such IP. While companies and universities may be aligned in the shared goal of creating technological innovations that benefit society, their interests may be misaligned in other respects, due to their different ultimate missions, as described above.

This chapter will delve into various aspects of IP in UI collaborations. It will explore the types of collaborations where IP is a key factor, the different kinds of IP that can emerge from these partnerships, and the factors influencing IP management. Additionally, it will examine the interests of both parties and the potential conflicts that may arise. Finally, the chapter will discuss strategies for mitigating conflicts between universities and industry regarding IP.

15.2 Opportunities and Challenges

For all the fundamental differences between universities and industry, there is a strong symbiotic relationship between those two sectors that has driven societal benefit and economic growth (Pressman et al., 2022). This has been particularly true since the passage of the Bayh-Dole Act in 1980 (Loise & Stevens, 2010), under which universities and other recipients of federal research funding in the United States are now able to retain title to inventions made with use of that funding. (Previously, inventions developed using federal funding remained the property of the US government, which had a very poor track record of bringing such inventions to the marketplace.) But why do universities and industry engage in research collaborations in the first place?

For universities, there are multiple benefits of UI collaboration, relating to funding, research access, commercial pathways, and bridging the gap between academic interests and market realities. First, although university research is predominantly funded by the federal government, the relative percentage of academic research expenditures from federal funding sources has been steadily decreasing, from about 70% in 1991 to under 60% 30 years later (AUTM, 2024). Industry, therefore, provides an alternative source of research funding, accounting for more than \$7.1 billion in the United States alone in 2023 (AUTM, 2024). Second, UI collaborations allow university researchers to access industry resources (such as specialized equipment, rapid prototyping, or large-scale manufacturing) that may not exist on the university campus, opening up experimental pathways that would otherwise be unavailable.

Third, UI collaborations provide a pathway to moving academic innovations from the bench to the market, which is the goal of many university innovators. This has the added benefit of providing faculty and student recruitment and retention advantages for the university, as these UI collaborations provide a real avenue for societal impact from these academic innovations. Working with industry from the outset of a research project allows an innovation to meet the industry's primary endpoints, rather than being developed in isolation and relying on the hope that it meets the needs of a future industry partner. Pursuing these UI collaborations also

allows for real-time feedback regarding the innovation from end-users, which is often not a possibility solely within the university's walls. And finally, depending on the terms of the agreement, successful commercialization of UI collaboration results may create additional financial upside for the university partner, sharing in the commercial success of the industry partner.

For industry, likewise, there are numerous benefits of UI collaboration. In some cases, it is the university that has expensive and specialized equipment and resources (such as electron microscopes, clean rooms, and test facilities) that would otherwise not be available to an industry partner. Additionally, industry can access unique and targeted expertise at the university around an area of technical interest. University researchers may have decades of expertise in a particular scientific area that could not be readily developed or replicated within a company. In addition, universities provide a consistent flow of experienced and motivated students who may become recruitment targets for companies.

In some cases, industry may even be able to "piggyback" on research funded by others, benefiting from the outcomes of projects supported by other entities (such as federal funding agencies or nonprofit foundations). This can be particularly advantageous for smaller companies or those with limited research budgets. It is important to note that universities have to tread carefully in this area, particularly with respect to federal research funding, state support, or local tax exemptions, to avoid the perception that public resources are being inappropriately used to support private interests.

An ideal UI collaboration results in a mutually beneficial scenario. University researchers can conduct publishable work that may not have been feasible otherwise, while industry gains exclusive access to technology or innovation it could not have developed independently. This collaboration can lead to the successful commercialization of a product based on the developed technology, benefiting the industry partner, the university, and society at large.

To maximize the likelihood of such a mutually beneficial outcome, the IP terms in UI collaboration agreements must be carefully crafted. These terms should be acceptable to both parties while also preserving the potential for the creation and protection of valuable IP. By striking the right balance between these objectives, universities and industry can foster collaborations that drive societal benefits.

15.2.1 Types of UI Collaborations and IP Considerations

UI collaborations can take various forms, each with distinct implications for IP ownership and management. These arrangements can range from one-off projects to long-term strategic partnerships.

One-off projects can include unfunded collaborations, where each party contributes to its own side of the work through independent funding sources; or jointly funded collaborations, where a university and company apply to a funding source together. Another one-off project option is a federal (or other third-party) flowdown arrangement, where a third-party entity, such as a government agency, provides

funding to either the university or the company, with some of the funding sub-awarded to the other party. In funded collaborations, the funding source may dictate IP terms vis-à-vis itself and the university and industry partner(s), which will then serve as a starting point for IP terms between the university and the industry partner(s).

Sponsored research, where a company provides funding from its own internal sources to a university for specific research projects, may also be a one-off project. Universities generally have default IP provisions for sponsored research agreements, which may need to be modified to meet the industry sponsor's needs. One-off projects may also be funded by gifts from companies to universities, supporting research with minimal IP "strings" (e.g., obligations for the university to offer rights in, or even disclose, IP developed using gift funding to the donor).

Beyond one-off projects, universities and industry can engage in more extensive and long-term collaborations. These partnerships can include elements beyond research, such as student recruitment/internship programs or personnel exchanges. These strategic partnerships often involve a master agreement outlining terms for multiple projects over a period of time. Ideally, with a master agreement in place, IP terms do not need to be renegotiated for each individual project. However, as the obligations of the IP terms of the master agreement will likely bind all future innovations, great care must be taken in drafting those IP terms for the benefit of both parties.

Consortium agreements extend beyond bilateral arrangements, involving multiple universities and/or multiple companies working together on a shared goal. These agreements can provide more expansive opportunities for resource sharing, knowledge exchange, and collaborative problem-solving. As discussed in further detail below, consortia that involve multiple industry participants, likely to be commercial competitors, will typically focus on "pre-competitive" research whose results are intended to be widely disseminated, with minimal IP protections.

Other arrangements may include joint appointments of faculty between universities and industry, students or postdocs seconded to companies, and even companies establishing a physical presence on university campuses. These arrangements can foster deeper integration and facilitate knowledge transfer. The IP provisions for these arrangements are generally covered by "Visiting Scientist" agreements that detail which institution would own any resulting IP: the home institution of the visiting scientist, the institution at which the visiting scientist is currently residing, or both (joint ownership).

Regardless of the specific arrangement between a university and an industry partner, certain IP-related issues are common across most types of collaboration. Understanding these issues is crucial for navigating IP rights and ensuring mutually beneficial agreements. Agreement terms need to cover IP ownership, procedures for IP protection, and rights in IP by the nonowning party. Most UI collaborations draw a distinction between IP generated during the course of the UI collaboration (foreground IP, or FIP) and IP developed independently from the UI collaboration (background IP, or BIP). The rest of this section addresses these issues, definitions, and solutions. The focus is primarily on IP provisions in UI collaborations involving only partners based in the United States. For international collaborations, it is

important to recognize that rules and laws regarding IP ownership and licensing vary by jurisdiction (WIPO, 2024), as do “cultural” expectations about UI relationships in general and IP in particular (see Chap. 17, International Research Collaborations).

15.2.2 The Challenges

Universities and industry each have their own interests and concerns regarding IP arising out of UI collaborations—and these interests may often be in opposition to each other.

Universities generally have a strong publication imperative, which can conflict with the goal of preserving IP rights. In a “publish or perish” environment, the ability of a university to disseminate its research results timely is sacrosanct, but such publication has the potential effect of inadvertently creating a patentability bar. While delaying or withholding publication may be possible for patentable inventions, it is often impractical or undesirable for the university and its researchers, particularly graduate students working on thesis projects.

Another key university interest is avoiding encumbrances on university IP. Universities typically seek flexibility in how they can manage and commercialize their IP, and they may be reluctant to agree to terms that restrict their options. As an example, faculty who are not involved in a specific UI collaboration are unlikely to agree to their IP being encumbered by the provisions of that collaboration.

In the United States, most universities are reluctant to assign outright to another party any IP developed by their researchers. This is a norm driven in part by concerns about the possibility of federal funding having been used in part to develop the IP in question. Federal funding regulations prohibit the assignment of said IP to a third party without explicit permission of the relevant federal funding agency. Generally speaking, a US university will offer, at most, an exclusive license to any of its IP.

Finally, universities expect fair remuneration if their IP results in a commercially successful product. This means that the university should be compensated commensurately for the value of its IP that contributed to the product’s success.

Industry partners also have specific interests regarding IP in UI collaborations. These interests often focus on ensuring the commercial viability and competitiveness of the developed technology. One key industry concern is freedom to operate (FTO). If valuable FIP arises from the UI collaboration, industry partners want to ensure that the university’s existing IP rights (BIP) will not hinder their ability to commercialize the technology. Accordingly, industry will often seek some kind of guarantee of free or minimal-cost access to any BIP controlled by its university partner. Another important consideration is exclusivity. Industry partners generally seek some degree of exclusivity as a form of competitive advantage in the marketplace. Freely published or unprotected innovations can be easily used by competitors, reducing the potential commercial value. Finally, industry partners are

concerned about fair pricing for IP licenses. They want to avoid being overcharged for access to technology they have helped to develop.

15.2.3 Conflict and Complications

The IP challenges in UI relationships lie in the inherent contradictions between the various interests described above. For instance, publication and academic norms, which emphasize sharing materials and knowledge, can conflict with industry's desire for IP protection and exclusivity. While an industry partner might be interested in trade secret protection, or a guarantee of exclusivity for unpatented or unpatentable materials, this is generally not an option in a university setting. Universities cannot keep their researchers from sharing their expertise with those outside of academia. While trade secrets can be kept within a strict and “locked-down” company environment, universities' inherent “leakiness” and expectation of being able to publish make pursuing trade secret protection for university innovations virtually impossible.

Offering the FTO that industry wants (specifically, free access to broadly defined BIP) would generally create unwelcome encumbrances for the university. From the industry perspective, a company would want access to, in addition to FIP developed during the UI collaboration, any other university IP necessary to provide FTO for that FIP. Unfortunately, that is incredibly challenging for universities to commit to at many levels. One university challenge to BIP concerns is due to the diffused research environment across a university, department, or even individual laboratories. This lack of centralization means that universities cannot be certain that they are aware of all potentially relevant BIP across the campus, or whether that BIP is already encumbered. This is further complicated by the fact that a UI collaboration generally benefits a particular laboratory with funding and resources, yet requested BIP may come from other laboratories that are not affiliated with the UI collaboration, potentially causing significant friction within the university.

Financial considerations add another layer of complexity, as universities typically aim to capture the full value for IP they transfer, while industry seeks to minimize its costs. Universities that develop FIP want to make sure that they have realized fair value for the innovations developed under the UI collaboration. A key, yet often misunderstood, tenet of this university perspective is that the funding received for the UI collaboration is meant to cover the costs of the research, not to pay for any subsequent inventions developed. A major concern here is that, since innovation cannot be predicted, any resulting invention from the UI collaboration may have significant value that is not fully realized by the institution. The industry partner, meanwhile, may feel like it has already paid for the research, so should own (or get a free, exclusive license to) any resulting FIP; or while it may recognize that value of an innovation can be determined only after conception, it worries that the university can “hold it hostage” and/or license the FIP to a third party, if license terms are not agreed to in advance.

Beyond these core issues, several additional factors must be addressed in IP provisions for UI collaborations. The involvement of students, whether on campus or at the company, necessitates careful structuring of IP and confidentiality agreements; students may or may not be subject to the same IP and confidentiality policies as employees (faculty, postdocs, staff scientists). Private Business Use (PBU) and tax implications must also be factored into the equation; for example, pre-negotiation of IP license terms may result in sponsored research activity being considered PBU, only a limited amount of which may be done at a university while remaining within the “safe harbor” provisions of the tax code. Public universities, in particular, need to take care to not be perceived as engaging in private commercial activity, but even private institutions may have buildings funded with public bonds that come with limitations on PBU. Universities need to ensure that IP terms for industry partners do not run afoul of export control regulations or jeopardize the fundamental research exclusion. Government rights, particularly in UI collaborations involving pass-through entities or direct federal funding, must be carefully navigated. Moreover, obligations to third parties, such as subawardees, contract research organizations, and additional collaborators, must be addressed. Finally, multiparty consortia introduce further complexities into IP arrangements.

The exact nature of the potential conflicts around IP in UI collaborations varies by industry sector, reflecting the broader variation in how IP is handled in different sectors. In pharmaceuticals and biotechnology, for example, the exclusivity offered by patent protection is paramount; a single composition of matter patent covering a novel therapeutic can be immensely valuable to its owner or licensee. Conversely, in the tech sector, innovation can be so rapid that patents cannot keep up with technical advances in a way that provides significant competitive advantage; instead, tech companies find themselves navigating through “patent thickets” (due to the large number of patents that may cover any given device, such as a smartphone) and may be most concerned about obtaining FTO.

15.3 Options, Solutions, and Impact

How can the challenges identified above be addressed, avoided, or resolved? A critical first step is clarity of definitions. What is considered to be IP under the relevant UI agreement, and hence subject to the agreement’s IP terms? How is ownership of IP generated under the agreement determined? What falls within the scope of FIP and BIP? Once those fundamentals are established, provisions can be negotiated that reflect a balance of all parties’ interests.

Note that while, in principle, all parties to a UI agreement (both university and industry partners) may be involved in generating IP, most of the considerations and concerns discussed here have to do with access of the industry partner(s) to IP generated by the university partner(s). Universities will rarely need rights in industry-controlled IP, except to the extent needed to perform under the UI agreement. Most UI agreements will include a simple statement confirming that each party has the

right to practice IP controlled by each other party as needed in the conduct of the research or other activity in question during the term of the agreement. Therefore, the remainder of this section focuses primarily on university IP to which the industry partner wants access for commercial purposes, with a mention of the special case of jointly owned IP.

15.3.1 Defining IP, FIP, and BIP

At one extreme, IP subject to the provisions of a UI agreement can be defined very narrowly, including only elements protected under IP law: inventions covered by patents, and software covered by copyright. A more expansive definition could encompass “inventions whether or not patentable” and “any material that is or may be copyrightable.” A very broad definition may extend to products of research activity that are “IP-adjacent,” such as data and other research results, technical specifications, process descriptions, design drawings, bills of materials, and even unwritten know-how. Many of these latter items will be exchanged as a matter of course as part of the UI collaboration and may not need special contractual arrangements. (Data and datasets are a rapidly evolving area, particularly in the context of machine learning and artificial intelligence, and are covered separately in Chap. 16, Data—Access, Generation, Use.) Here is a possible definition of IP:

“Intellectual Property (IP)” means Inventions and Copyrightable Material, where “Invention” means any invention or discovery that is or may be patentable or otherwise protectable under Title 35 of the United States Code, and “Copyrightable Material” means any material or other property, including computer software, that is or may be copyrightable or otherwise protectable under Title 17 of the United States Code.

Once the parties have agreed on what is encompassed by “IP,” the scope of FIP and BIP has to be established. The FIP definition will rarely be controversial: it will include IP (however defined) generated in the conduct of the Scope of Work (SoW) of the UI agreement, during the term of the agreement, typically limited to IP developed in the lab(s) of the university principal investigator(s) (PI(s)) named in the agreement. For one-off agreements, the SoW and term should be well defined; more extensive agreements may include sub-agreements with individually defined SoWs. Here is a possible definition of FIP:

“Foreground IP (FIP)” means IP [as previously defined] that is invented or authored in the performance of the project described in the Statement of Work, under this Agreement, (a) by University personnel solely in the laboratory of the Principal Investigator(s) named in this Agreement; (b) by Company personnel; or (c) jointly by such University and Company personnel.

In conjunction with such a definition, the agreement should specify how ownership of the FIP is to be determined. In many cases (especially in the United States), this will simply be “ownership follows inventorship”: each author or inventor assigns rights to his or her employer, such that a piece of IP with all university authors/

inventors is owned solely by the university, a piece of IP with all company authors/inventors is owned solely by the company, and a piece of IP with a mix of such authors/inventors is jointly owned by the university and the company. However, there may be additional considerations in play having to do with the location of authors/inventors at the time of creation, and the use of either party's resources. Many universities and many companies require that visitors to their respective institutions, and users of their respective equipment and other resources, sign agreements under which those visitors assign IP rights to the university/company being visited or whose resources are being used, overriding those visitors' obligations to their home institutions. When such visits or resource sharing take place in the context of a UI collaboration, the agreement governing that collaboration should clearly spell out how IP ownership is to be treated in those circumstances. This needs to be framed particularly carefully in situations where a given researcher may have a contractual relationship with both the university and the industry partner: for example, a student conducting an internship at the company, or a company scientist with an academic appointment at the university.

BIP provisions, in contrast, are more likely to be heavily negotiated; this will include discussions of both the scope of BIP as well as the rights that the industry partner receives in the university's BIP, however defined. As discussed above, the industry partner in a UI agreement will want sufficient FTO to be able to exploit the FIP developed in the collaboration; this may even extend to the extreme of asking the university partner to indemnify it against the infringement of third-party IP—a provision to which virtually no university will agree. Unfortunately, even a more reasonable (from the industry perspective) request for FTO limited to only BIP controlled by the university will be difficult for the university to accommodate. The university, meanwhile, may wish the agreement to remain silent on the topic of BIP, completely decoupling the licensing of BIP rights from the terms of the UI agreement—a position that the industry partner is unlikely to see as being in the spirit of collaboration.

In practice, a middle ground can usually be found using one or more restrictions on the scope of BIP. The parties may endeavor to identify BIP existing as of the effective date of the agreement and explicitly list it in the agreement; but the unpredictability of research and the possible range of FIP and other project results or deliverables makes it inherently unlikely that all possible BIP can be so identified. BIP may be restricted temporally, ideally to IP existing as of the effective date, or perhaps through the term of the UI agreement. A common compromise on scope is to limit BIP to that developed in the laboratory of the PI(s) funded by, or collaborating under, the UI agreement. Finally, given that FTO with respect to FIP is the primary concern with respect to BIP, the latter should be defined as IP that would necessarily be infringed by performance of the FIP, or, for copyright, IP that is incorporated into the FIP. Language that addresses this point by restricting BIP to IP that is "necessary" for the exploitation of the FIP or other results of the research collaboration is commonly proposed, but needs to be used cautiously: if the research results constitute a small component of a large system, BIP should not include IP

that covers other components of that same system not directly related to the UI research project at hand.

At a minimum, a university should insist on language that exempts IP that is already encumbered from the definition of BIP. Otherwise, it may find itself in an untenable position: with conflicting commitments to different parties over the same IP.

The above provisos regarding BIP apply primarily to patents; a different set of considerations come into play when the FIP, and hence the potential BIP, is software. Much “new” software incorporates previously existing code, which may be covered under any of a number of open source or other licenses. Depending on the license in question, the incorporation of this existing code (BIP) into new code (FIP) may automatically make the FIP subject to whatever open-source provisions apply to the BIP. For software-related projects, the parties should discuss expectations regarding the use of open-source code, libraries, and other software components, and if necessary, set limits on the use of such components, or require prior notice of such use. Unlike inventions and patents, for software it will generally be possible to identify likely BIP prior to starting the project.

15.3.2 Conveyance of IP Rights

In addition to defining IP, FIP, and BIP, the parties must also agree on the terms under which rights to university-owned FIP and BIP are conveyed to the industry partner. These terms may depend on the type of UI agreement under which the collaboration is taking place. At one extreme, a gift agreement should include no rights in university IP back to the donor company (“no strings”). At the other extreme, a company may enter into a technical services agreement (the scope of which should be very narrow, and should not include research; for example, analysis of company-provided components using specialized electron microscopy techniques available only at the university) under which the company owns (or receives a royalty-free exclusive license to) all rights to results (e.g., data or images resulting from the analysis) and IP generated by the university in the performance of the agreement (generally with the exception of IP pertaining to improvements in analysis techniques developed incidentally in the course of performing the contracted services). Most UI collaborations are somewhere in the middle.

Generally speaking, when an industry partner funds research at a university, it will automatically receive a non-exclusive, royalty free (NERF) license to FIP, for internal or noncommercial use only (i.e., the company can make and use, but not sell, items covered by the university’s patents on FIP or incorporating university software that is FIP). The agreement will generally also include a time-limited (e.g., 6 months) option to negotiate a royalty-bearing, commercial, nonexclusive or exclusive license to FIP. In some cases, a university may grant a commercial NERF license to FIP under the original agreement. Terms for BIP rights vary, but

(particularly if the definition of BIP is of reasonably limited scope), these may be included in the FIP license at no additional cost to the industry partner.

While the automatic NERF license to FIP is quite common, it can be beneficial to the university to require its industry partner to affirmatively elect such a license, and to make it clear in the terms of the UI agreement that in the absence of that election, no such license is granted. This allows the university to exclusively license that same FIP to a different company, should the original industry partner not be interested, without requiring a “but for” carveout in the third-party license. Importantly, in the United States, most, if not all, universities qualify for discounted (“small entity”) fees at the United States Patent and Trademark Office (USPTO); most industry partners are “large entities” and do not qualify for this discount, and the grant of any patent rights to a large entity precludes the university from paying discounted fees with respect to the patent rights in question.

The internal-use NERF license to FIP, and option for greater rights, may be expressed as follows:

Company will receive a nonexclusive, worldwide, royalty-free license in the Field [as defined elsewhere] to use University FIP for its own internal purposes. This nonexclusive license will not include the right to sublicense and is nontransferable, except in the event of the transfer or sale of all or substantially all of Company’s assets relating to the Field to a third party.

Company will receive a time-limited option to negotiate an exclusive or non-exclusive commercial, worldwide, royalty-bearing license in the Field to University FIP, subject to University’s IP policy and any prior agreements University may have with other sponsors. If Company wishes to negotiate a license, it shall notify University in writing within six (6) months following University’s written disclosure of University FIP (“Option Period”). If Company timely notifies University that it wishes to negotiate a license, University and Company agree to negotiate in good faith and execute a license agreement within three (3) months after Company’s notice (“Negotiation Period”), on the same terms as University would provide to an unrelated, non-sponsoring party; *i.e.*, a competitive price determined at the time the licensed technology is available for use.

In any commercial license granted herein, University will reserve for itself and other non-profit research institutions a royalty-free, worldwide, irrevocable license to make and use University FIP for research, educational and clinical research purposes. If applicable, any such license will also provide that the rights of the United States of America as set forth in Public Laws 96-517 and 98-620 (codified at 35 U.S.C. § 200 *et seq.*) are especially reserved, and that Company shall comply with the provisions of 35 U.S.C. § 204, as amended from time to time.

If Company does not exercise its option above to negotiate a license during the Option Period, or if the parties fail to execute a license agreement prior to the end of the Negotiation Period, then University may freely exploit or license its interests in the University FIP without further obligation to Company.

To address the concerns industry may have about being “held hostage” in the negotiation of financial terms for the license to FIP developed using its funding or other resources, universities have developed alternative pricing approaches, often offering their industry partners a menu of options. This may include a one-time “tech fee” for rights to any FIP developed under the agreement, either per invention disclosure (or patent filing elected by the industry partner, to avoid incentivizing the university to generate excessive numbers of invention disclosures), or per agreement. In the

latter case, the university may get less overall for a productive project with many invention disclosures but may also get “something for nothing” if the project does not yield any disclosures. The parties may also include a “bonanza clause” where the industry partner pays an additional amount to the university upon reaching a certain sales threshold for goods covered by the licensed university IP; this protects the university against the risk of “giving away the store” if unusually valuable FIP is developed in the collaboration.

For example, the University of Minnesota’s “MN-IP Create” program (University of Minnesota, [n.d.](#)) offers industry partners three options with respect to FIP (defined as “all inventions arising from the research project”). Under Option A, companies pay the greater of 10% of the sponsored research agreement value or \$15,000 for an exclusive, sublicensable license to FIP, with no annual minimums, milestone payments, or other fees, but with a 1% royalty when net sales (of products using the licensed FIP) exceed \$20 million a year. Option B is the “standard” terms described above: a noncommercial NERF license, with a 6-month option to negotiate a royalty-bearing license to FIP. And under Option C, companies pay the greater of 10% of the sponsored research agreement value or \$10,000 for a commercial NERF license to FIP, with an option to negotiate a royalty-bearing exclusive license. Under all these options, BIP is “not automatically included, but the University will work closely with the sponsoring company to resolve any requests.”

IP provisions in UI agreements should also clearly specify the mechanisms for disclosure of FIP between the parties; how the parties will coordinate patent filing, prosecution, and maintenance; and responsibility for the expenses associated with the latter. Agreements should also cover the handling of jointly owned IP. Under United States patent law, a joint owner’s rights in a jointly owned patent are essentially equivalent to a commercial NERF license; typically, the agreement will include in the option and licensing provisions not just the university’s solely owned FIP, but also the university’s rights in any jointly owned FIP.

Any agreements governing UI collaborations that are expected to yield patentable subject matter need to address the tension between the academic pressure to publish, and the public disclosure bars to patentability. This is generally addressed by explicitly stating that the parties anticipate publishing the results of the work, while allowing for a short (1 or 2 month) delay to allow review of proposed publications for patentable subject matter, and the filing of patent applications should any such subject matter be found. (This review period can also be used to check for, and remove as necessary, any confidential information of either party that may have found its way into the draft publication.) The following language may be used to cover publication provisions:

University may release any information or publish any material resulting from the project. University shall furnish Company with a copy of any such proposed publication preferably thirty (30) days prior to, but in no case later than simultaneously with, submission for publication for Company to identify Company Confidential Information or patentable inventions. Company has thirty (30) days to provide written notification to University of Company Confidential Information disclosed in a proposed publication. Should such Company Confidential Information be identified, University agrees to revise the publication

to comply with the confidentiality obligations of this Agreement. Upon written notification by Company of an invention disclosed in a proposed publication, University agrees to use reasonable efforts to file a U.S. patent application sufficient in University's sole judgment to preserve patent rights in the invention. University shall have no obligation to revise or delay publication to accommodate such a U.S. patent application.

Of course, the above terms, while typical of UI sponsored research agreements in the United States, are not the only possible IP arrangements for UI collaborations. In scenarios where university and industry personnel are working particularly closely together, with use of both university and company resources, the parties may agree that all FIP should be jointly owned, regardless of details of inventorship or authorship. In other cases, the parties may agree that the research being conducted is "pre-competitive" and explicitly forgo any IP protection, agreeing to put inventions into the public domain by not filing patent applications, and to make copyrighted software available under an open-source license; as discussed previously, this is seen perhaps most often in multi-party consortium agreements (more so in the tech sector than in the life sciences), where companies that are competitors in the marketplace nevertheless work together with each other and with academia to develop new science/technology to advance their field more generally. And, outside of the United States, university researchers may have the right to retain ownership in the IP they develop; this needs to be taken into account when universities wish to offer IP rights to their industry partners in UI collaborations.

15.3.3 Internal Communication

When entering into a UI collaboration, whether it be a simple one-off sponsored research agreement or a long-term strategic partnership, it is critical that university staff negotiating the terms of the agreement clearly communicate those terms to the PIs and other researchers involved in the collaboration. While the university PIs are typically heavily involved in developing the SoW and budget for the research to be conducted, and may be in close communication with their scientific counterparts at the industry partner regarding technical aspects of the proposed work, negotiation of the IP terms generally falls to the university's Technology Transfer office, Sponsored Programs office, and/or General Counsel's office. Provisions with which researchers need to be familiar (and may wish to weigh in on during the negotiations) include the scope, disclosure process, and industry partner rights in FIP and BIP; and the industry partner's right to review, and request (limited) changes to, proposed publications. At a minimum, any PIs who may be involved in a UI collaboration should be informed of their obligations under the proposed agreement, and given the opportunity to decline participation if the terms are not to their liking. Some PIs may wish to be involved at the negotiation stage, and university staff conducting the negotiations should run proposed IP and publication terms by such PIs prior to executing an agreement with the industry partner. Other PIs may not be familiar with the typical IP terms in UI agreements, and may need to be educated

about the potential implications of such terms for them and their research teams. It is particularly important to make it clear to entrepreneurial PIs that the IP provisions of a UI collaboration could encumber IP from their laboratories in a way that may complicate any future startup activity relating to the technology in question. Postdocs, students, and other research personnel in the laboratories involved in UI work should be put on notice of IP terms that would pertain to their own research (or their own plans to start a company). Graduate students and postdocs, in particular, should never be put in a position where their ability to freely and timely publish their research is constrained.

15.4 Takeaways and Future Strategies

As seen in this chapter, IP can be both the most valuable and the most contentious aspect of a UI collaboration. With proper planning, however, university and industry partners can come to an agreement on IP provisions that meet the needs of all parties.

First, the parties need to clearly set expectations regarding IP and identify any potential conflicts upfront. Is protectable IP a desired outcome of the collaborative work? Are industry partners looking for exclusive rights in project IP as a source of competitive advantage in the marketplace, or is the collaborative work intended to be pre-competitive, fundamental research that advances the knowledge base in a particular field? Are companies aware of the university's need to own the IP arising from its researchers' work, and to be able to freely publish research results?

Once expectations have been aligned, the parties need to agree on agreement language that accurately conforms to those expectations. This should include clear definitions or descriptions of

- The types of IP covered by the agreement
- Foreground and background IP (FIP and BIP)
- Rights granted by university to company in BIP, if any
- Terms under which the company gets rights in FIP
- Processes and procedures for managing project IP

UIDP and its member organizations from academia and industry have put considerable effort into working together to document university and industry perspectives, considerations, and principles regarding IP in UI collaborations in a set of Contract Accords, including for software (UIDP, 2020a), copyright (UIDP, 2020b), BIP (UIDP, 2020c), and FIP (UIDP, 2020d). These can serve as useful additional resources in the negotiation process.

It is also important to include the right people from both sides at the negotiation table. UI collaborations are often initiated by researchers from the prospective collaborative organizations, but these individuals may not be aware of the organizational expectations and constraints that exist on both sides of the collaboration. Ideally, this will include individuals who understand both the business imperatives

and the legal and policy constraints for each party. On the university side, this may include personnel from the Sponsored Projects office, Technology Transfer (especially for the IP terms discussed in this chapter), and possibly General Counsel. On the industry side, the primary negotiators will be an Academic Alliances or similarly positioned team, with support from the General Counsel's office as necessary. For UI collaborations that go beyond one-off projects, such as strategic partnerships or consortium agreements, universities may have dedicated Corporate Relations or Corporate Partnerships teams whose role is to manage the overall relationship between the university and its industry partners and guide agreement negotiations (including over IP) to a mutually beneficial outcome.

With the right people at the table, expectations properly managed, and language carefully crafted, IP considerations should enhance—and not be an impediment to—a successful UI collaboration.

References

- AUTM. (2024). *AUTM 2023 licensing activity survey*. <https://autm.net/surveys-and-tools/surveys/licensing-survey/2023-licensing-survey>
- Loise, V., & Stevens, A. J. (2010). The Bayh-Dole Act turns 30. *Science Translational Medicine*, 2(52) <https://www.science.org/doi/10.1126/scitranslmed.3001481>
- Pressman, L., Planting, M., Moylan, C., & Bond, J. (2022). *Economic contributions of university/nonprofit inventions in the United States: 1996–2020*. BIO and AUTM. https://autm.net/AUTM/media/About-Tech-Transfer/Documents/BIO-AUTM-Economic-Contributions-of-University-Nonprofit-Inventions_14JUN2022.pdf
- UIDP. (2020a). *Contract accord 16: Software*. <https://uidp.org/publication/contract-accord-16-software/>
- UIDP. (2020b). *Contract accord 8: Copyrights*. <https://uidp.org/publication/contract-accord-8-copyrights/>
- UIDP. (2020c). *Contract accord 5: Background intellectual property*. <https://uidp.org/publication/contract-accord-5-background-intellectual-property/>
- UIDP. (2020d). *Contract accord 6: Foreground intellectual property*. <https://uidp.org/publication/contract-accord-6-foreground-intellectual-property/>
- University of Minnesota. (n.d.). *MN-IP: Sponsoring research & innovation partnerships*. <https://research.umn.edu/units/techcomm/corporate-engagement/sponsoring-research-mn-ip>
- Wilson, M. B. (2008). Definitions of intellectual property. In *AUTM technology transfer practice manual*, Vol. 3, Part 3: https://www.autm.net/AUTMMain/media/ThirdEditionPDFs/V3/TTP_V3_P3_IP.pdf
- WIPO. (2024). *Study on patent inventorship and ownership issues arising from collaborative research and cross-border collaboration*. WIPO. https://www.wipo.int/edocs/mdocs/scp/en/scp_36/scp_36_9.pdf

Stephen J. Susalka is the Chief Executive Officer of AUTM, a 3,000+ member non-profit association focused on supporting and enhancing technology transfer globally. He also oversees a cooperative agreement with the Federal Laboratory Consortium for Technology Transfer to support technology transfer for more than 300 federal labs across the US. As an internationally recognized leader in technology commercialization, Susalka leverages his extensive experience in

fostering successful university-industry collaborations, along with his expertise in IP, licensing, and startups, to empower AUTM members in bringing groundbreaking innovations to market.

Hannah Dvorak Carbone is the Director for Innovation, Patents & Licensing in the Office of Technology Transfer and Corporate Partnerships (OTTCP) at the California Institute of Technology (Caltech), where she has worked since 2007. She earned a PhD in biology at Caltech, is a registered US patent agent, and served on AUTM's Board of Directors from 2020 to 2022.